Assessing the Value of Data-Driven Discrete-Event Simulation for Emergency Department Operations: Insights from Large-Scale Multicenter Hospital Data

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Topic: Data-driven strategies to mitigate overcrowding and improve operational efficiency in emergency departments

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PURPOSE

Emergency Departments (EDs) operate as socio-technical systems characterised by stochastic demand and heterogeneous clinical pathways (Hybinette et al., 2025). These features make EDs vulnerable to overcrowding, which negatively affects quality of care, patient safety and efficiency.

Discrete-Event Simulation (DES) has emerged as a widely adopted analytical approach to study ED operations, enabling the representation of patient-level flows, queues, resources utilisation and time-dependent decision processes (Vecillas Martin et al., 2025).

Despite the growing availability of datasets, most DES applications continue to be grounded in single-hospital case studies and simplified downstream representations. This raises concerns about the robustness and transferability of DES models when applied across heterogeneous hospitals, limiting their suitability for system-level decision making (Zhang, 2018).

This limitation is manifest in the Italian context, where evidence on multi-site DES applications remains scarce. Existing contributions rely on single-site simulations and heterogeneous modelling assumptions, which constrain cross-study comparability and generalisability of findings across organisational settings (Tedesco et al., 2023).

METHODOLOGY

The study adopts a retrospective observational design, based on the analysis of emergency department administrative data referring to the year 2023. The dataset includes fourteen EDs from seven hospitals located in Northern Italy. All healthcare organizations provided data through standardised Case Report Forms (CRFs).

The collected data included anonymised demographic, clinical, temporal and organisational variables. Data were harmonised and processed using Python.

To capture patient trajectories beyond the ED, ED records were integrated with hospital discharge data, enabling estimation of inpatient length of stay (LOS) and modelling of downstream capacity constraints and boarding. In addition, the National Emergency Department Overcrowding Score (NEDOCS) (Weiss et al., 2004) was computed on an hourly basis to characterise overcrowding dynamics.

Based on these empirical data, data-driven DES model was developed to represent the main operational processes of the participating EDs, including triage, clinical assessment, discharge and interaction with hospital bed availability. Model input parameters were derived directly from the administrative data, including arrival patterns, acuity-specific process times, routing probabilities between care stages, resource and capacity constraints such as treatment spaces, staffing and downstream bed availability. Stochastic inputs will be represented using empirically derived distributions.

The approach follows a two-step logic: first, descriptive analyses will be conducted to characterise patient flow dynamics and overcrowding patterns; second, these empirical results will be used to parameterise the DES models, which will be employed to reproduce observed system behaviour and to explore alternative organisational scenarios.

FINDINGS

The empirical analysis identifies a structural congestion across the fourteen participating EDs. Across sites, the system operates close to or above its effective capacity, with 35–55% of observed hours spent in overcrowded or severe overcrowding states (NEDOCS > 100), generating systematic delays not attributable to demand surges.

High-acuity patients systematically exceed the 8-hour target. Average LOS is 11:28:25 for Yellow patients and 11:14:52 for Red patients, remains well above target also for Orange patients (09:45:22). In contrast, lower-acuity patients experience shorter total LOS (around 3–6 hours for lower-acuity patients), but with disproportionately long waiting times. All high-acuity categories exceed the 8-hour target by 1.7–3.5 hours on average, indicating that congestion originates beyond initial clinical assessment.

Temporal patterns of arrivals and NEDOCS confirm that overcrowding persist beyond peak hours. Similar arrival volumes yield different congestion levels, indicating state-dependent capacity effects.

By integrating ED and hospital discharge data, downstream bed availability emerges as the dominant driver of these dynamics. Periods of high boarding are tightly aligned with prolonged LOS for admitted patients, who are predominantly high acuity, explaining why their LOS exceeds target values even when clinical priority rules are respected.

The data-driven DES models reproduce these regularities only when downstream capacity constraints are explicitly represented, generating prolonged LOS for admitted patients, queue accumulation for low-acuity cases and recurrent transitions into overcrowded regimes, based on Monte Carlo simulation experiments with 100,000 simulated trajectories per hour.

RELEVANCE

This study provides empirical evidence on the feasibility and practical value of data-driven DES in the context of Italian EDs, a setting where simulation tools remain underutilized. By leveraging routinely collected administrative data from several hospitals, the proposed approach demonstrates how DES can be grounded in real-world operational conditions and used to support system-level decision-making across hospital-wide patient flows.

From a managerial and policy perspective, the findings highlight the potential of DES to enhance understanding of overcrowding dynamics, identify structural inefficiencies, and support proactive planning strategies. In particular, the results show that ED overcrowding is an endogenous property of hospital-wide patient flow rather than a consequence of isolated demand shocks, and that DES provides a robust basis for evaluating organisational interventions in tightly coupled healthcare networks. The study positions DES as a key enabler of digital transformation in emergency care, bridging the

gap between data availability, analytical capability, and operational decision-making in complex socio-technical healthcare systems (Zhang., 2018).

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